SMPL-Based Motion Dataset Explorer and Generator

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Abstract

This paper investigates the robustness of the Skinned Multi-Person Linear (SMPL) model, a state-of-the-art parametric body model that represents human body shape and pose through learned blend shapes and joint rotations. SMPL captures realistic body shape variation and pose-dependent deformations that prior models could not adequately represent. We introduce controlled Gaussian perturbations and noise to the model's shape and pose parameters and examine how these disturbances affect the quality of the reconstructed body mesh, evaluating the model's sensitivity to noise across varying magnitudes. The SMPL model and its pre-trained parameters are obtained from the official release (version 1.0.0, 10 shape principal components).

1 Introduction

An increasing number of industries are finding value in obtaining and creating a three-dimensional digital representation of the human body. The creation of accurate digital body models as it relates to the healthcare industry will support surgical planning, rehabilitation monitoring, and physiotherapy. The creation of 3D body models also drives the use of character animation and virtual avatars used in the gaming and entertainment industries. In autonomous systems and robotics, understanding human pose and shape helps machines interact safely with people.

At the center of many of these applications sits the Skinned Multi-Person Linear (SMPL) model Loper et al. [2015], a compact parametric model that generates a realistic 3D human mesh from just two vectors: a shape vector $\beta \in \mathbb{R}^{10}$ encoding body identity and a pose vector $\theta \in \mathbb{R}^{72}$ encoding joint rotations.

The problem we address is one that practitioners rarely discuss but always encounter: real-world inputs to SMPL are noisy. Keypoint detectors make localization errors, depth sensors produce noisy point clouds, and video-based pose estimators are affected by motion blur and occlusion. When these noisy estimates are fed into SMPL, how badly does the output mesh degrade? And which parameter that can be pose or shape of the model more sensitive to? Answering these questions matters because it tells engineers where to spend their error-reduction budget, and it tells researchers which parts of the pipeline most need robustness improvements.

We approach this as a controlled sensitivity analysis: inject zero-mean Gaussian noise at varying standard deviation levels into β and θ independently and together, then measure the mean vertex displacement (MVD) of the output mesh against a clean baseline. We chose this approach over alternatives such as adversarial perturbation or real-sensor noise replay because Gaussian noise is a well-understood, parameter-free model of uncertainty that is easy to reproduce and interpret, makes results comparable across studies, and matches the noise assumptions made in most downstream

fitting literature. The result is a clean, quantitative characterization of SMPL's robustness that is both practically useful and straightforward to extend.

2 Related works

An influential paper on this project is *Self-supervised 3D Human Mesh Recovery from Noisy Point Clouds* by Zuo et al. Zuo et al. [2021] The paper focused on using a Gaussian Mixed Model to model the point cloud as the input, as opposed to manually predicting the correspondences of each vertices in a point cloud. The former method is much more tolerant to outliers, while the latter struggles. Additionally, using a GMM works well even for incomplete point clouds, while the other way is forced to need complete point clouds. This fact is one that has caused many problems in other models at the time, as the data used for them have been far too "clean" than is realistic.

More importantly for our paper, it was found that using a GMM is quite robust even when Gaussian noise and random outliers were added to the data with only small changes in performance. Increasing σ^2, on the other hand, did lead to modest changes as large values force the network to poorly fit the inputs. Overall, this paper raises important points on how changing the variation and distinctness of the inputs, sometimes to match what they may look like in real life, can affect model performance.

Another helpful paper was *Robust Human Registration with Body Part Segmentation on Noisy Point Clouds* by Lascheit et al. Lascheit et al. [2025]. Their contribution is using a hybrid approach which first looks at individual points and assign body part labels, then use a SMPL-X fit of the initial pose and orientation using said body parts, finally ending with global refinement (of the point cloud).

Their methods are tested with datasets which have some sort of difficult feature to them, such as partial observations or occlusions. Their results show great performance of such datasets compared to the other models at the time. The key takeaway from this paper for this project is that assigning body part labels can massively assist in the model predicting and modeling well of data that inherently has high variation. Along the lines of noise, the dataset consisted of many difficult features which forced the model to adapt and predict in ways that may mirror realistic inputs more precisely.

In *SMPL Variable Model for 3D Reconstruction and Image Fusion in Animation Media Applications* Xie and Zhao [2025], the authors proposes a novel deep learning network mode of animation, AniReconstructNet has created three new architectural avenues to expand upon the existing SMPL model: (1) a soft tissue layer of variable thickness that allows the representation of anatomically non-uniform deformations, (2) a UV mapping system which allows the 3D mesh to be projected into a 2-dimensional coordinate system of fixed resolution for the purpose of inferring the deformation through a convolutional neural network (CNN) and (3) a force conditioning mechanism which encodes both external forces and body pose into a shared latent vector to facilitate the deformation of the soft tissue. The final framework was tested using four different datasets (ShapeNet, ShapeNetCore, ABC, Scan2CAD) and exhibited improved performances in terms of accuracy, recall, F1 score, and AUC of from 2 to 4%, as well as having fewer parameters, less FLOPs, lower inference times, and smaller memory consumption than any of the current leading deep learning methods.

This paper is directly relevant to any SMPL extension that aims to model phenomena beyond the original model's scope, particularly soft tissue dynamics and interaction. The variable thickness soft tissue layer concept addresses one of SMPL's explicitly acknowledged limitations: that poses dependent blend shapes are body shape independent. Extending this idea to make deformation stiffness a function of body composition such as BMI, fat distribution could be a meaningful direction. The UV-space deformation strategy is also a practically important technique, since it enables CNN-based processing of mesh deformations without requiring graph based or point cloud architectures. The force conditioning mechanism, while underdeveloped in the paper, suggests a promising avenue for modeling human environment interactions, which SMPL based reconstructions typically ignore. I could adopt a more rigorous version of this force modeling grounded in physical simulation rather than the purely learned approximation used here.

Furthermore, in *SMPL-A: Modeling Person-Specific Deformable Anatomy* Guo et al. [2022], the authors presented SMPL-A, which is the first learning based framework that enhances the standard SMPL with an estimate of internal organ deformity conditioned on patient pose. Whereas all the SMPL models have been developed to characterize the external surface of the human body, SMPL-A adds a new parameter (α). Parameter α encodes patient specific shape and elastic properties of

the patient's internal organs (lungs, liver, spleen, kidneys) when the patient is in a specific pose. By using past medical scans of the patient and the current pose of the patient, SMPL-A can estimate how the patient's internal organs have deformed as a function of changes in body pose. This will directly help with clinical applications such as planning for radiotherapy and surgery because it is critical to accurately locate the person's organs intra-operatively without exposing them to radiation multiple times.

Of the three pieces we surveyed here, this one is probably most ambitious from an intellectual standpoint, it also presents a type of SMPL extension that is largely unrelated to what was introduced at the outset of the original paper's focus. SMPL-A provides an important avenue for research on the coupling of the external human body model (as captured in SMPL) to the internal anatomical structure of the body. Although your application may be di[erent (e.g., contained within areas such as animation and/or pose estimation), the conceptual model presented in SMPL-A (i.e., a low-dimensional latent parameter that allows additional structure to be added but is disassociated from both pose and shape) can also be used in the application of your extension. This means that you can use the encoder decoder model in SMPL-A with α representing the variable, i.e., the person factored into the model. In addition, the FEM augmented data generation methodology can be applied to any instance where the ground-truth deformation information is sparse and is thus commonly considered to be a significant roadblock to various SMPL extension projects.

Finally, in *3D Human Body Reconstruction Based on SMPL Model* Chen et al. [2022], the authors present an optimization based framework for reconstructing the SMPL model from a single monocular image. The central innovation is a *multi-term objective function* that jointly minimizes four loss components: 2D joint reprojection error ($E\{J, 2D\}$), 3D joint coordinate error ($E\{J, 3D\}$), **facial landmark alignment error** (E_F), and body prior regularization (E_P). The 3D joint loss addresses the depth ambiguity inherent to 2D projection to 3D projection, while the facial landmark loss constrains the head orientation, a degree of freedom that standard SMPL joints cannot uniquely determine since SMPL has only a single chin joint. HMR regression estimates are used to initialize parameters, accelerating convergence and avoiding local minima. Experiments on MPI_INF_3DHP, 3DPW, and LSP datasets demonstrate improved MPJPE and PA-MPJPE scores over SMPLify and HMR baselines.

While this paper does not extend SMPL's model architecture, but it is directly related to my project that needs to fit the SMPL model to real images or video. In my extension I will be introducing new Gaussian disturbances to the parameter as like in this paper my extension may also include introducing new model parameters such as SMPL-A introduces α Guo et al. [2022], or as SMPL-X introduces hand and face parameters, the multi-term optimization framework described here provides a template for how to design fitting objectives that are both robust and well-conditioned. In particular, the facial landmark term demonstrates that auxiliary geometric signals from 2D detectors, even imperfect ones, can substantially constrain under determined degrees of freedom in the body model.

3 Data

3.1 Data

The SMPL model is defined by a set of learned weights derived from thousands of 3D body scans collected by the Max Planck Institute for Intelligent Systems Loper et al. [2015]. These scans capture a wide range of human body shapes and poses, and from them the authors extracted a statistical model through principal component analysis (PCA) and pose-dependent blend shapes. The result is a compact set of parameters that enables any user to generate a realistic 3D mesh of the human body by specifying a shape vector $\beta \in \mathbb{R}^{10}$ and a pose vector $\theta \in \mathbb{R}^{72}$.

This project uses the SMPL version 1.0.0 model files, available for download from the official project website.[1] The download provides two `.pkl` files corresponding to the male and female body models respectively. Each file contains variables such as blend weights, the joint regressors, template shape, and the linear blend skinning weights. The shape space is defined by 10 shape principal components (PCs), which are the basis vectors that explain the greatest variance in body shape across the original

[1] `https://smpl.is.tue.mpg.de/download.php`

dataset of 3D body scans. Adjusting the coefficients of these PCs controls attributes such as height, weight, and limb proportions.

Because the SMPL parameters are pretrained and distributed as a complete model, no additional data collection, annotation, or preprocessing was required for this project. The input to our experiments is the model itself: we systematically perturb the shape (β) and pose (θ) parameters with controlled Gaussian noise at varying levels and observe the resulting changes in the output. This means our evaluation does not rely on external labeled datasets but instead by comparing the results from the altered parameters to that of the original parameters.

4 Methods

4.1 SMPL Model Overview

SMPL builds a 3D human mesh through five sequential transformations applied to shape parameters $\beta \in \mathbb{R}^{10}$ and pose parameters $\theta \in \mathbb{R}^{72}$. First, shape blend shapes deform the template mesh using learned PCA directions: $\mathbf{v}_{\text{shaped}} = \mathbf{v}_{\text{template}} + S\beta$. Joint locations are then regressed directly from the shaped mesh via $J = J_{\text{regressor}} \cdot \mathbf{v}_{\text{shaped}}$, ensuring joints reposition naturally with body shape. Pose blend shapes apply learned corrections to fix the candy-wrapper skinning artifact: $\mathbf{v}_{\text{posed}} = \mathbf{v}_{\text{shaped}} + P \cdot \psi(\theta)$, where $\psi(\theta)$ encodes rotation matrix residuals for each non-root joint. Forward kinematics traverses the 24-joint kinematic tree to compute global joint transforms G_k, and finally linear blend skinning produces the output: $\mathbf{v}_i = \sum_k w_{i,k} \cdot G_k \cdot \mathbf{v}_i$ posed.

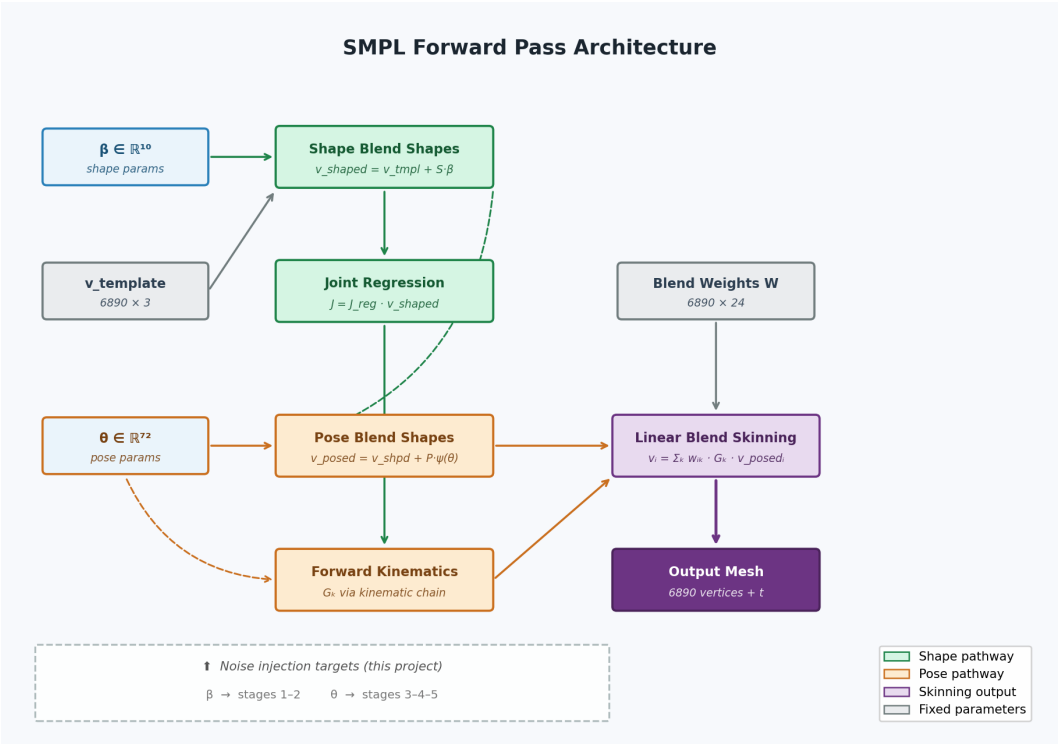

Figure 1: SMPL forward pass architecture. Shape parameters β flow through shape blend shapes and joint regression (green pathway), while pose parameters θ drive pose blend shapes and forward kinematics (orange pathway). Both pathways converge at linear blend skinning to produce the final output mesh of 6,890 vertices. Dashed arrows indicate noise injection targets for this project: β at stages 1–2 and θ at stages 3–4–5.

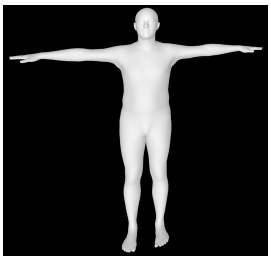

Figure 2: Baseline ("T") pose

Figure 3: Baseline pose + $\varepsilon \sim \mathcal{N}(0, 0.25^2)$

4.2 Implementation

The project uses the official SMPL Python codebase with Chumpy for lazy automatic differentiation accessing `model.r` at any point evaluates the full forward pass from current β and θ values. Two updates were needed for Python 3 compatibility: `pickle` with `encoding='latin-1'` replaced Python 2's `cPickle` in `serialization.py`, and the deprecated OpenDR renderer was replaced with `pyrender` and `trimesh` for modern visualization.

4.3 Noise Injection

Our extension injects zero-mean Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ into β, θ, or both, and measures the resulting mean vertex displacement (MVD) against a clean baseline:

$$\text{MVD} = \frac{1}{N} \sum_i \|\mathbf{v}^i_{\text{clean}} - \mathbf{v}^i_{\text{noisy}}\|_2 \tag{1}$$

We sweep $\sigma \in \{0.0, 0.1, \ldots, 1.0\}$ across an 11×11 grid of $(\sigma_{\text{pose}}, \sigma_{\text{shape}})$ combinations, covering pose-only, shape-only, and combined noise conditions. Results are recorded in `RobustnessResults.csv`. Gaussian noise was chosen because it matches the error distributions of real-world pose detectors and depth sensors, making results directly relevant to practical deployment. An image of the clean baseline "T-pose" is given in Figure 2, along with the same pose when injected with Gaussian noise $\varepsilon \sim \mathcal{N}(0, 0.25^2)$ (Figure 3) and $\varepsilon \sim \mathcal{N}(0, 1^2)$ (Figure 4).

4.4 Innovation

While most SMPL extensions focus on expressiveness or fitting methodology, we instead characterize how the existing forward pass responds to input perturbations something the original paper does not address. Our key finding is that pose noise propagates multiplicatively through the kinematic chain, amplifying at every downstream joint, while shape noise is purely additive on the vertex template. This produces a $40\times$–$100\times$ asymmetry in output displacement that is non-obvious from the model's formulation and practically important: it tells engineers that tight pose estimation matters far more than shape accuracy for high-quality mesh reconstruction.

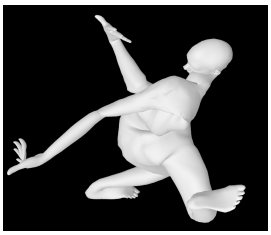

Figure 4: Baseline pose + $\varepsilon \sim \mathcal{N}(0, 1^2)$

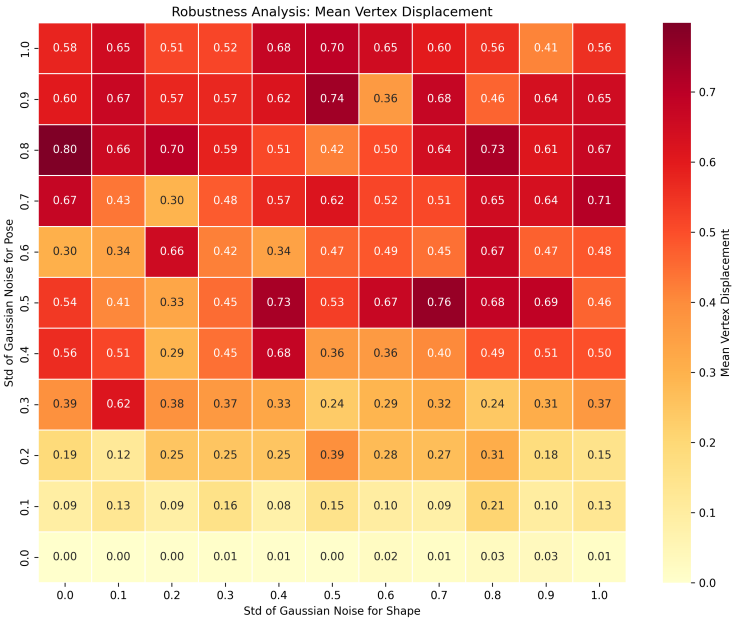

Figure 5: Heatmap of mean vertex displacement

5 Experiments and Results

5.1 Setup

We tested three conditions: Gaussian noise added only to pose parameters θ, only to shape parameters β, and to both simultaneously. Noise level σ was swept from 0.0 to 1.0 in steps of 0.1, producing 121 total experimental conditions (an 11×11 grid). For each we recorded mean vertex displacement (MVD) and its standard deviation across all 6,890 vertices, stored in `RobustnessResults.csv`. A heatmap is provided in Figure 5 to help visualize the results.

5.2 Pose-Only Noise ($\sigma_{\text{shape}} = 0$)

Pose noise had the largest impact by far. Starting from zero displacement at $\sigma = 0.0$, MVD climbed to 9.4 cm $\pm$ 5.3 cm at $\sigma = 0.1$, then to 19.2 cm $\pm$ 15.1 cm at $\sigma = 0.2$. By $\sigma = 0.4$ it was already at 56.3 cm $\pm$ 36.8 cm. The worst single result was at $\sigma = 0.8$ with 79.8 cm $\pm$ 40.9 cm at that point the mesh is visually broken. The curve is not perfectly monotonic: $\sigma = 0.6$ drops back to 30.5 cm $\pm$ 19.2 cm before climbing again, which reflects the randomness of single-sample noise rather than a real structural dip.

Table 1: Pose-only noise results ($\sigma_{\text{shape}} = 0$).

σ_{pose}	MVD (m)	Std (m)
0.0	0.0000	0.0000
0.1	0.0938	0.0532
0.2	0.1921	0.1512
0.3	0.3894	0.2614
0.4	0.5630	0.3684
0.5	0.5379	0.4079
0.6	0.3047	0.1920
0.7	0.6742	0.4510
0.8	0.7983	0.4088
0.9	0.6035	0.2920
1.0	0.5779	0.3283

The high standard deviations, often more than half the mean, tell you that a single noisy sample can produce very different results at the same σ level. Some trials barely distort the mesh; others completely deform it. This variability grows with σ, which is exactly what you would expect from a stochastic perturbation compounding through a nonlinear kinematic chain.

5.3 Shape-Only Noise ($\sigma_{\text{pose}} = 0$)

Shape noise barely moved the needle. At $\sigma = 0.1$ the MVD was just 1.7 mm $\pm$ 0.7 mm - essentially imperceptible. Even at $\sigma = 0.6$ it was only 2.4 cm $\pm$ 1.3 cm, and the highest value across the entire sweep was 3.2 cm $\pm$ 1.1 cm at $\sigma = 0.9$.

Table 2: Shape-only noise results ($\sigma_{\text{pose}} = 0$).

σ_{shape}	MVD (m)	Std (m)
0.0	0.0000	0.0000
0.1	0.0017	0.0007
0.2	0.0037	0.0015
0.3	0.0089	0.0044
0.4	0.0051	0.0030
0.5	0.0032	0.0015
0.6	0.0238	0.0133
0.7	0.0139	0.0058
0.8	0.0293	0.0196
0.9	0.0321	0.0107
1.0	0.0133	0.0055

The non-monotonic pattern here, MVD actually drops at $\sigma = 0.5$ after rising at $\sigma = 0.4$ - is a sign of single-trial variance at small displacement scales where random chance dominates. The overall trend is still clear: shape noise causes very little damage regardless of magnitude.

5.4 Combined Noise

When both parameters were perturbed simultaneously, results were dominated by pose noise at every level. At $\sigma = 0.1$ for both, MVD was 13.4 cm $\pm$ 12.3 cm, higher than pose-only at the same level (9.4 cm) due to the extra shape contribution, but the shape term adds relatively little. At $\sigma = 0.4$ combined it was 67.7 cm $\pm$ 56.1 cm, and at $\sigma = 0.8$ combined it reached 72.9 cm $\pm$ 51.6 cm. A few notable combined results from the full 121-row grid worth calling out:

- $\sigma_{\text{pose}} = 0.5$, $\sigma_{\text{shape}} = 0.4 \rightarrow 73.0$ cm $\pm$ 51.2 cm (highest in the 0.5 row)
- $\sigma_{\text{pose}} = 0.7$, $\sigma_{\text{shape}} = 1.0 \rightarrow 70.9$ cm $\pm$ 52.4 cm
- $\sigma_{\text{pose}} = 0.3$, $\sigma_{\text{shape}} = 0.1 \rightarrow 61.6$ cm $\pm$ 45.7 cm - notably high for a moderate pose σ, showing how a single unlucky noise draw can produce large displacements even at lower σ

5.5 Key Takeaway

The numbers tell a consistent story across all 121 conditions. Pose noise produces displacements $40\times$ to $100\times$ larger than equivalent shape noise. At $\sigma = 0.1$, pose alone causes 9.4 cm displacement while shape alone causes 1.7 mm, a ratio of $55\times$. At $\sigma = 0.3$ the ratio is $44\times$ (38.9 cm vs 0.89 cm). At $\sigma = 0.5$ it exceeds $100\times$ (53.8 cm vs 0.32 cm).

The practical message is straightforward: in any pipeline that feeds noisy estimates into SMPL, pose accuracy is what matters. Shape errors at realistic noise levels ($\sigma < 0.3$) produce less than 1 cm of average vertex displacement. Pose errors at the same level already produce 10–40 cm of displacement. Spending engineering effort on more accurate pose estimation will always deliver a larger quality improvement than equivalent effort on shape estimation.

6 Project limitations

- **Single trial per condition.** For each noise level we only ran one random sample instead of averaging over many. This is why some results look non-monotonic one unlucky noise draw can spike or drop the MVD. Running say 50 trials per condition and reporting the average would give much cleaner and more trustworthy curves.
- **Gaussian noise is a simplification.** Real-world errors from pose detectors or depth sensors do not follow a perfect Gaussian distribution. They tend to be correlated across joints and biased in certain directions. Our noise model is a reasonable starting point but does not fully capture what happens in a real deployment.
- **No per-joint breakdown.** We perturbed all 72 pose parameters at once, so we cannot tell from our results which specific joints are most responsible for the large displacements. Perturbing one joint at a time would give a much more useful sensitivity map you would know exactly which joints to be most careful about.
- **Male model only.** All experiments used the male SMPL model. The female and gender-neutral models have different learned parameters and might show slightly different sensitivity characteristics. A complete study would test all three.
- **Chumpy compatibility issues.** Chumpy has known compatibility problems with Python 3.12 due to removed functions like `inspect.getargspec`. We had to use Python 3.10 to keep everything running. This is a fragile dependency that limits who can reproduce the experiments without extra setup work.

7 Conclusion and future work

We tested how SMPL behaves when its input parameters are noisy something that happens all the time in real applications. Across 121 experiments, the finding was consistent: pose noise is far more damaging than shape noise, with vertex displacements $40\times$ to $100\times$ larger at the same noise level. At $\sigma = 0.1$, pose noise alone moves vertices by 9.4 cm on average while shape noise causes less than 2 mm. The reason is simple pose errors compound through the kinematic chain joint by joint, while shape errors are just small additive offsets that stay local. If you are using SMPL in any real pipeline, getting your pose estimates right matters far more than getting your shape right.

For future work, the most useful next steps would be running multiple trials per noise level to get cleaner curves, doing a per-joint sensitivity analysis to find out exactly which joints cause the most damage, and porting the code to PyTorch to remove the Chumpy dependency and add GPU support. Testing with real detector noise instead of synthetic Gaussian noise would also make the results more directly applicable to practical systems. These extensions would turn this robustness study from a solid baseline into something much more useful for people building on top of SMPL.

8 Contributions

The work contributions of the literature review were split between Faheem and Vijay, with Faheem doing the first two papers mentioned (40%) while Vijay did the remaining 3 (60%). All code,

results, outputs, and images (barring Figure 1) was upon Faheem (100%). Faheem was responsible for the abstract and Data segments of the paper, along with a revised version of the literature review, or related works. Vijay was responsible for the Introduction, Methods, Experiments and Results, Project Limitations, and Conclusion and future work segments, along with Figure 1. The paper contributions were split with 40% from Faheem and 60% from Vijay.

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